# URL-Based Phishing Attack Detection

SATURDAY THE 15/02/2025


WORK REALISED BY:

NEJMEDDINE BEN MAATOUG
HAMZA HENTATI
BACHAR TOUNSI




# Introduction

In this Report, we tried deep learning so we employed three deep learning models which are : **Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM)**, and a **Hybrid (CNN+LSTM)** model. These models were trained and evaluated on the **web phishing detection dataset** to classify websites as either phishing or legitimate

Deep learning was chosen for its ability to automatically extract features from raw data, making it highly effective for complex tasks like phishing detection. Below, we present the methodology, results, and analysis for each model.

# Dataset Analysis

The dataset used in this project consists of 11,429 URLs, each labeled as either phishing or legitimate. It includes 89 features extracted from the URLs and website content. These features include:

- URL-based features: Length of URL, presence of special characters, domain information, etc.
- Content-based features: Number of external links, use of iframes, SSL certificate status, etc.
- Domain-based features: Age of domain, DNS records, etc.

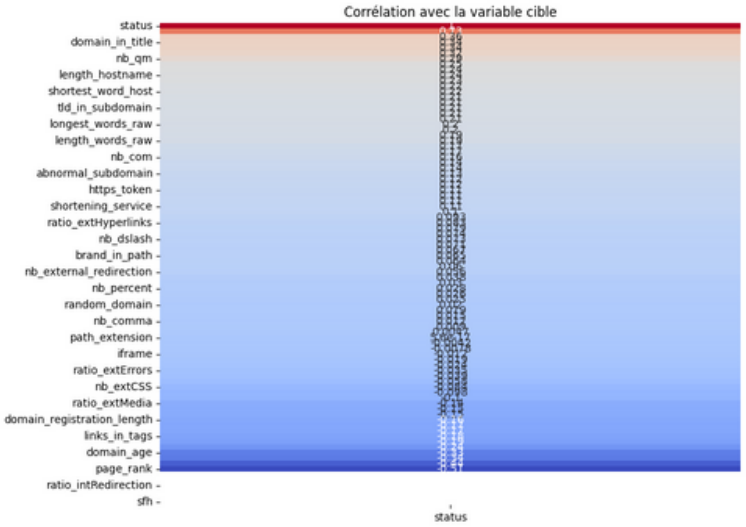
The dataset is balanced, with an equal number of phishing and legitimate samples, ensuring unbiased model training.

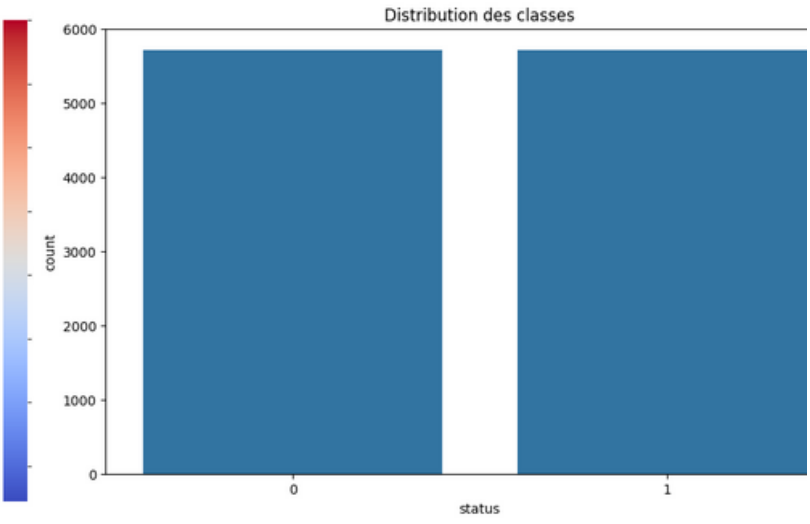
Dataset Statistics

- Total URLs: 11,429
- Number of Features: 89
- Phishing URLs: 5,714
- Legitimate URLs: 5,714

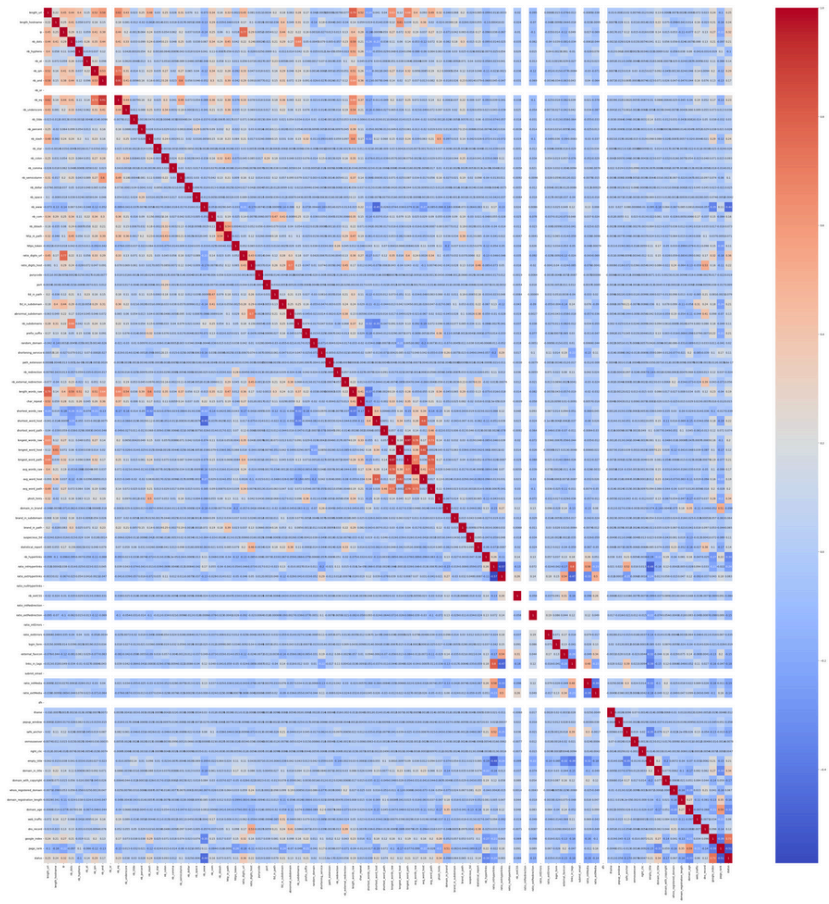
# Exploratory Data Analysis

- **Correlation Matrix**: The correlation matrix shows the relationships between features. Highly correlated features (e.g., URL length and number of special characters) were identified and retained for model training.
- **Class Distribution**: A bar plot shows the distribution of phishing and legitimate URLs, confirming the dataset is balanced.




- **Feature Matrix**: The feature matrix visualizes the distribution of feature values across the dataset. This helps in understanding the range and scale of each feature.



# Convolutional Neural Network (CNN)

**Overview:**

CNNs are primarily used for image processing but can also be applied to structured data like URLs and website features. In this project, we adapted a CNN to analyze the feature vectors extracted from the phishing dataset.

**Code Implementation:**

The CNN model was implemented using TensorFlow/Keras. The code includes:

- Data preprocessing: Normalization and reshaping of feature vectors.
- Model architecture: Convolutional layers, pooling layers, and fully connected layers.
- Training: The model was trained for 31 epochs with a batch size of 681,729 params.

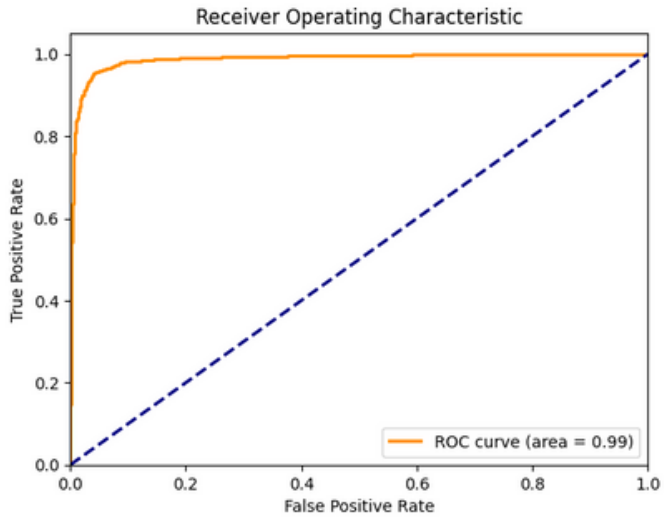
**Results:**

The CNN model achieved an very important values listed in the image below and a final accuracy of 95%

```
72/72 ━━━━━━━━━━━━━━━━━━━━ 1s 7ms/step
              precision    recall  f1-score   support

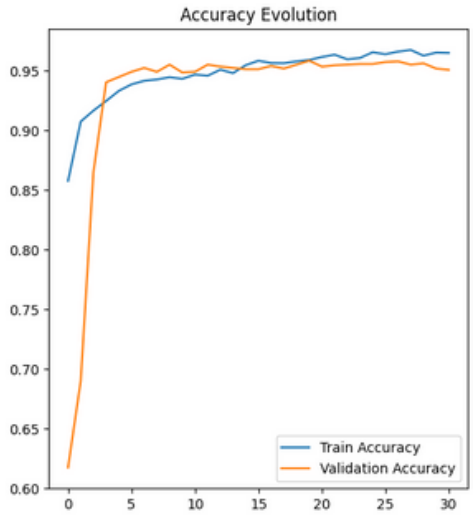
           0       0.96      0.94      0.95      1143
           1       0.94      0.96      0.95      1143

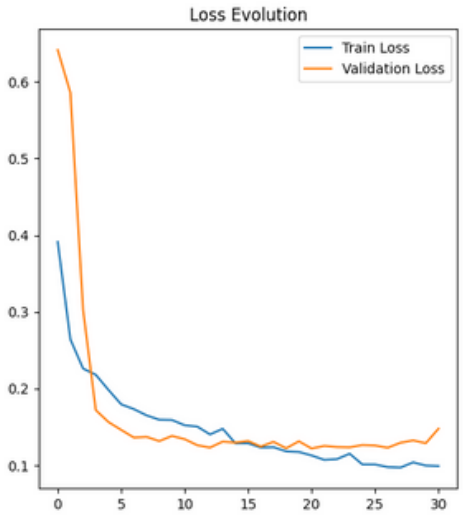
    accuracy                           0.95      2286
   macro avg       0.95      0.95      0.95      2286
weighted avg       0.95      0.95      0.95      2286
```




Link To The Code :

https://colab.research.google.com/drive/1IVajTVgqKuyorJL_psKNjxSygkWwK1fj



# Long Short-Term Memory (LSTM)

**Overview:**

LSTMs are a type of recurrent neural network (RNN) designed to handle sequential data. In phishing detection, LSTMs can capture temporal dependencies in URL structures or feature sequences.

**Code Implementation:**

The LSTM model was implemented using TensorFlow/Keras. The code includes:

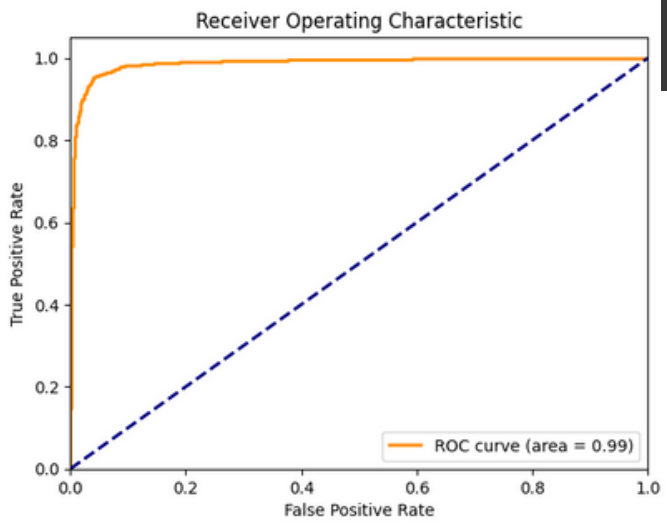
- Data preprocessing: Sequential feature extraction and padding.
- Model architecture: LSTM layers followed by dense layers.
- Training: The model was trained for 31 epochs with a batch size of 171,905 params.

**Results:**

The LSTM model achieved an very important values listed in the image below and a final accuracy of 96%

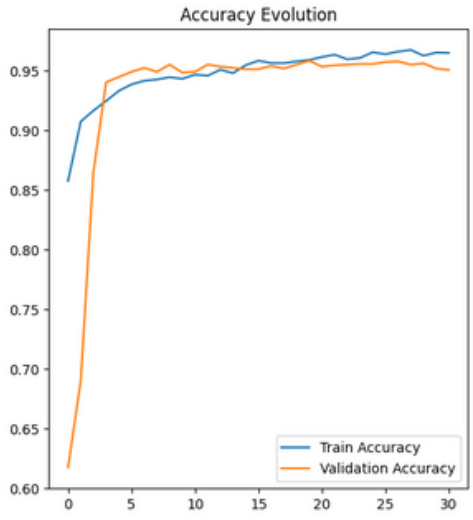
72/72 ━━━━━━━━━━━━━━━━━━━━ 3s 31ms/step

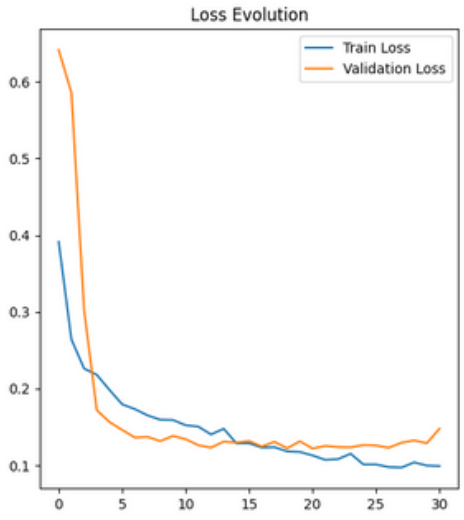
| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0 | 0.95 | 0.97 | 0.96 | 1143 |
| 1 | 0.96 | 0.95 | 0.96 | 1143 |
| accuracy | | | 0.96 | 2286 |
| macro avg | 0.96 | 0.96 | 0.96 | 2286 |
| weighted avg | 0.96 | 0.96 | 0.96 | 2286 |




Link To The Code :

https://colab.research.google.com/drive/1ZxR1uOpea5n1XtclCG0IgnEZwAfOMxDs#scrollTo=bseYDMDh6sXQ



# Hybrid CNN-LSTM Model

**Overview:**

The hybrid model combines the strengths of CNNs and LSTMs to leverage both spatial and temporal features. This approach is particularly effective for datasets with complex patterns.

**Code Implementation**

The hybrid model was implemented using TensorFlow/Keras. The code includes:

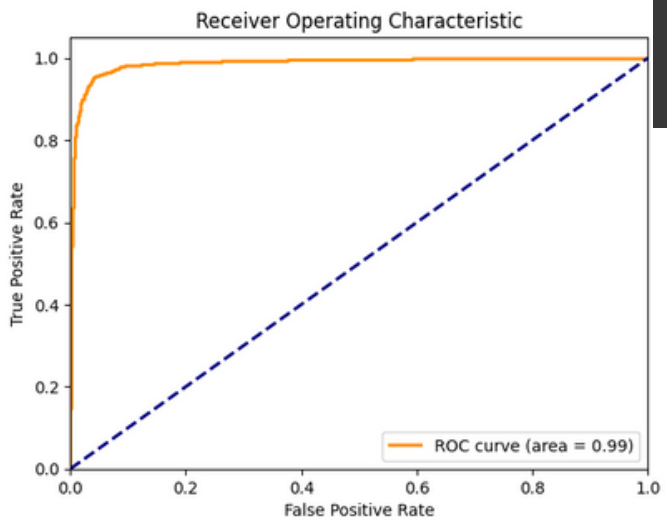
- Data preprocessing: Feature vector normalization and sequential data preparation.
- Model architecture: CNN layers for spatial feature extraction followed by LSTM layers for temporal dependencies.
- Training: The model was trained for 40 epochs with a batch size of 92,161 params.

**Results:**

The Hybrid model achieved a very important values listed in the image below and a final accuracy of 96%

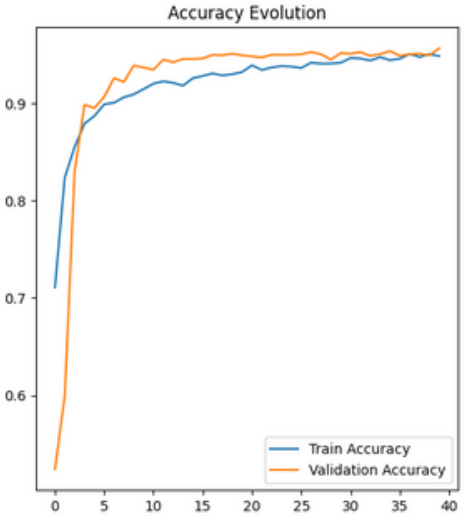
72/72 ━━━━━━━━━━━━━━━━━━━━ 1s 12ms/step

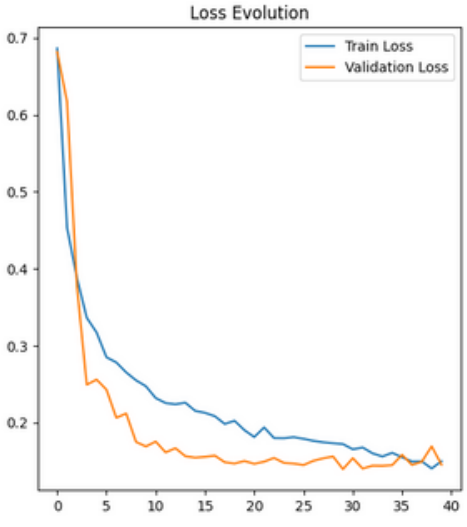
| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0 | 0.95 | 0.94 | 0.94 | 1143 |
| 1 | 0.94 | 0.95 | 0.94 | 1143 |
| accuracy | | | 0.94 | 2286 |
| macro avg | 0.94 | 0.94 | 0.94 | 2286 |
| weighted avg | 0.94 | 0.94 | 0.94 | 2286 |




Link To The Code :

https://colab.research.google.com/drive/1jyZluejvIcd-8SQi6nswA_dzpmG1_SeC#scrollTo=AS_5xJV92t4x



# Multi-Layer Perceptron (MLP)

**Overview:**

The MLP model is a powerful feedforward neural network that effectively captures complex patterns in textual data. By leveraging fully connected layers with non-linear activations, the MLP can learn meaningful representations for NLP tasks such as text classification and sentiment analysis.

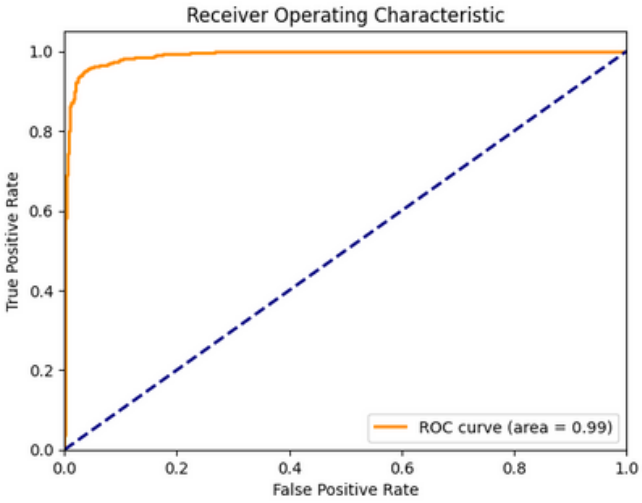
**Code Implementation**

The model was implemented using TensorFlow/Keras. The code includes:

- Data preprocessing: Tokenization, vectorization, and feature scaling to ensure optimal input representation.
- Model architecture: Multiple dense layers with ReLU activation, followed by a softmax layer for classification.
- Training: The model was trained for 100 epochs with a batch size of 128, optimizing with Adam and SGD and categorical cross-entropy loss.

**Results:**

The MLP model demonstrated strong performance, achieving high accuracy reaching 96%.

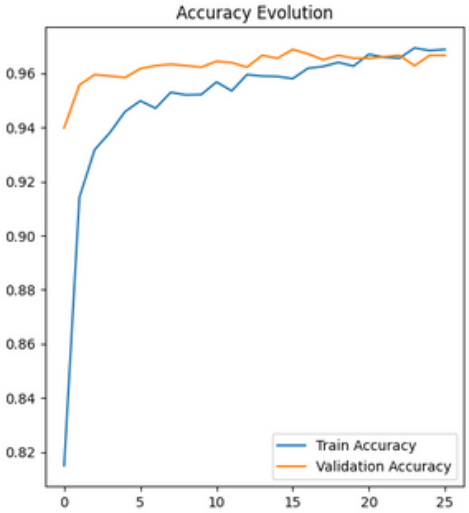


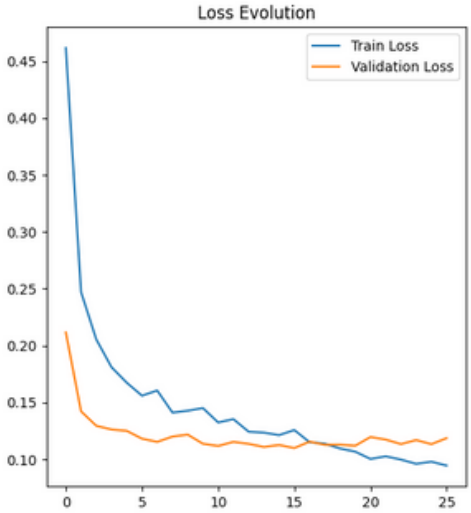

72/72 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| 0 | 0.95 | 0.97 | 0.96 | 1143 |
| 1 | 0.97 | 0.95 | 0.96 | 1143 |
| accuracy | | | 0.96 | 2286 |
| macro avg | 0.96 | 0.96 | 0.96 | 2286 |
| weighted avg | 0.96 | 0.96 | 0.96 | 2286 |

Link To The Code :

https://colab.research.google.com/drive/19kCwTG8ybtAlFGa160LzzslfhtzXMUKo#scrollTo=gHP5Dezys3Ok



# Comparative Table of our Tests

| Deep Learning Technique | Dataset | Parameters | Metric | Link To the code |
|---|---|---|---|---|
| **CNN using Adam (Convolutional Neural Network)** | Web Phishing Detection Dataset | Total params: 682,625<br>Trainable params: 681,729<br>Non-trainable params: 896 | Accuracy: 95%<br>Recall: 96%<br>F1-score: 95% | https://colab.research.google.com/drive/1IVajTVgqKuyorJL_psKNjxSygkWwK1fj |
| **CNN using SGD (Convolutional Neural Network)** | Web Phishing Detection Dataset | Total params: 682,625<br>Trainable params: 681,729<br>Non-trainable params: 896 | Accuracy: 95%<br>Recall: 95%<br>F1-score: 95% | https://colab.research.google.com/drive/1qhWLvWiaFF6FdW8hourgvDduQx3yaa0- |
| **LSTM using Adam (Long Short-Term Memory)** | Web Phishing Detection Dataset | Total params: 172,801<br>Trainable params: 171,905<br>Non-trainable params: 896 | Accuracy : 95%,<br>Recall : 95%<br>F1-score : 95% | https://colab.research.google.com/drive/1veuYzHjCNI4eGFQVEIcgDWLTMJxMSoVI |
| **LSTM using SGD (Long Short-Term Memory)** | Web Phishing Detection Dataset | Total params: 172,801<br>Trainable params: 171,905<br>Non-trainable params: 896 | Accuracy : 96%,<br>Recall : 96%<br>F1-score : 96% | https://colab.research.google.com/drive/1ZxR1uOpea5n1XtclCG0IgnEZwAfOMxDs#scrollTo=bseYDMDh6sXQ |
| **CNN + LSTM using Adam (Hybrid Model)** | Web Phishing Detection Dataset | Total params: 93,057<br>Trainable params: 92,161<br>Non-trainable params: 896 | Accuracy : 94%<br>Recall : 95%<br>F1-score: 94% | https://colab.research.google.com/drive/1jyZIuejvIcd-8SQi6nswA_dzpmG1_SeC#scrollTo=AS_5xJV92t4x |
| **CNN + LSTM using SGD (Hybrid Model)** | Web Phishing Detection Dataset | Total params: 93,057<br>Trainable params: 92,161<br>Non-trainable params: 896 | Accuracy : 94%<br>Recall : 95%<br>F1-score: 94% | https://colab.research.google.com/drive/1jeup5wDJpAbAx06Ctotoiv5k7mWtbJJq |
| **MLP using Adam** | Web Phishing Detection Dataset | Total params: 22,529<br>Trainable params: 22,801<br>Non-trainable params: 448 | Accuracy : 96%<br>Recall : 97%<br>F1-score: 96% | https://colab.research.google.com/drive/19kCwTG8ybtAlFGa160LzzslfhtzXMUKo#scrollTo=2YBH-7ZFtVwu |
| **MLP using SGD** | Web Phishing Detection Dataset | Total params: 22,529<br>Trainable params: 22,801<br>Non-trainable params: 448 | Accuracy : 95%<br>Recall : 94%<br>F1-score: 95% | https://colab.research.google.com/drive/10jyFj-H3gckr2KROv1SLNJ49QkVTyRWc#scrollTo=GsMuV70Z_4fj |